

How IoT can make the Workplace Intelligent

It's well-known that an intelligent workplace enhances employee productivity. The Internet of Things plays a big role in making a workplace smart and intelligent. Let's see how.



Enterprises demand a flexible, intelligent, and connected workplace as it enables its occupants to work smarter. It also helps in enhancing employee productivity and experience, leading to better employee retention and helping attract a new talent pool. Internet of Things (IoT) enables an intelligent office to utilise physical devices for collecting and sharing data. It helps enterprises to design flexible, activity-centric workspaces to better meet employees' evolving needs by analysing IoT sensors and connected system data on how space is being utilised. It enriches employee journey maps (such as seamless access, automatic room and desk booking, and colleague or way finding) from the moment they enter the office, and creates a conducive work environment by using smart temperature and light controls.

IoT for an intelligent workplace

An intelligent workplace uses automated processes to control an office's operation including heating, ventilation, air conditioning, lighting, security and other systems. A smart office uses sensors, actuators and microchips to collect data and manage it according to business functions and services. This infrastructure helps real estate and facility managers improve asset reliability and performance, which reduces energy use, optimises how space is used and minimises the environmental impact of buildings.

Heating, ventilation, and air conditioning (HVAC): Enterprises can digitally integrate an office HVAC system with sensors and software. They can provide mobile applications and smart technologies that allow employees to customise the temperature and lighting in their workspaces. This helps in:

- Air quality and temperature management based on occupant needs
- Automatic switch off of air conditioning and heating to avoid unnecessary energy use
- Control of humidity levels

Networked lighting: Digital integration of lighting fixtures with sensors and software can create an environment that maximises occupants' comfort and saves energy. Such lighting helps to:

- Adjust light to occupancy levels
- Control lighting as per the personal choice of occupants
- Save energy
- Adjust light based on ambient conditions, and control window shades

Occupancy sensors: These identify the utilisation of a workplace position through movement and/or thermal measurement. A sensor can be positioned at any location occupied by a person. It helps in monitoring:

- Office utilisation
- Desk utilisation
- Meeting room utilisation
- Heat maps to illustrate occupancy of office
- People counter – counts real-time occupancy
- Room sensor – detects if a room is being used or not

Access and security: This involves use of wearables (smartwatch, smart badge, etc) to authenticate employees for access to buildings, meeting rooms, equipment, devices and data. Smart wristbands could become ID badge replacements (or smart ID badges could be created) in the office space. For example, the building will recognise your car when you arrive, authenticate it, and then direct you to a parking